Efficient Bayesian inference in models for cosmological structure formation

Running MLDA on partially stochastic models

Louise Kluge[1], Linus Seelinger[2], Robert Scheichl[3], Dylan Nelson[4], Sebastian Krumscheid[1]

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[1]Scientific Computing Center, Karlsruhe Institute of Technology
[2]Pasteur Labs, NY 11205, USA
[3]Zentrum für Astronomy, ITA, Heidelberg University
[4]Institute for Mathematics and Interdisciplinary Center for Scientific Computing (IWR), Heidelberg University



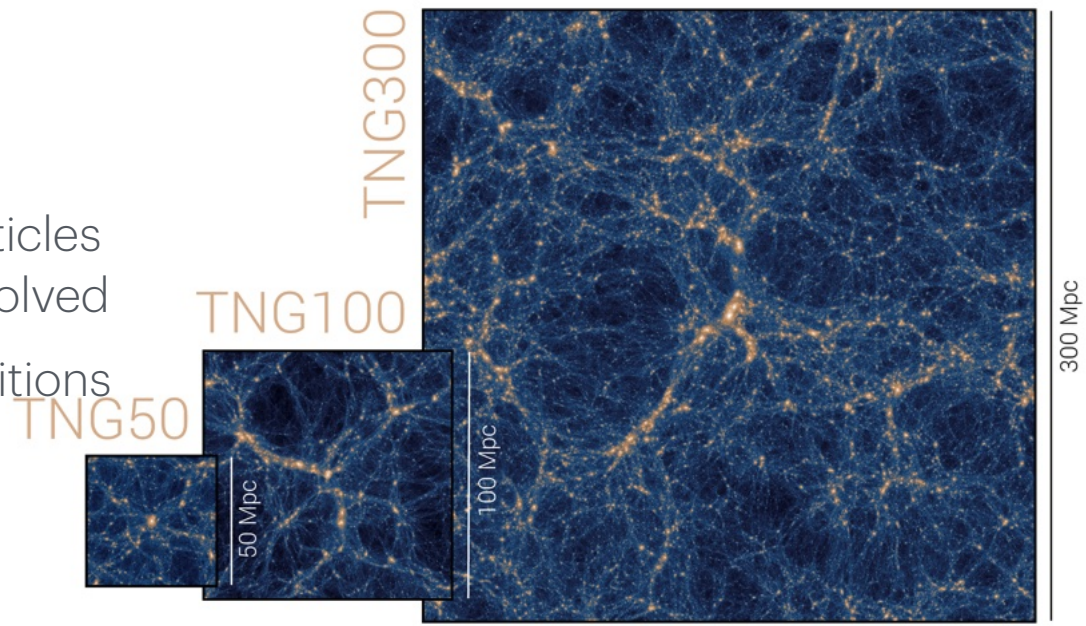


The IllustrisTNG Model

(Pillepich et al. 2017)

- cosmological simulations model large-scale matter distributions under
  - cosmic expansion
  - gravity
  - magnetohydrodynamics
- SPH-based, matter is discretised into particles for which gravity and hydrodynamics is solved
- large boxes with periodic boundary conditions
  → large runtime

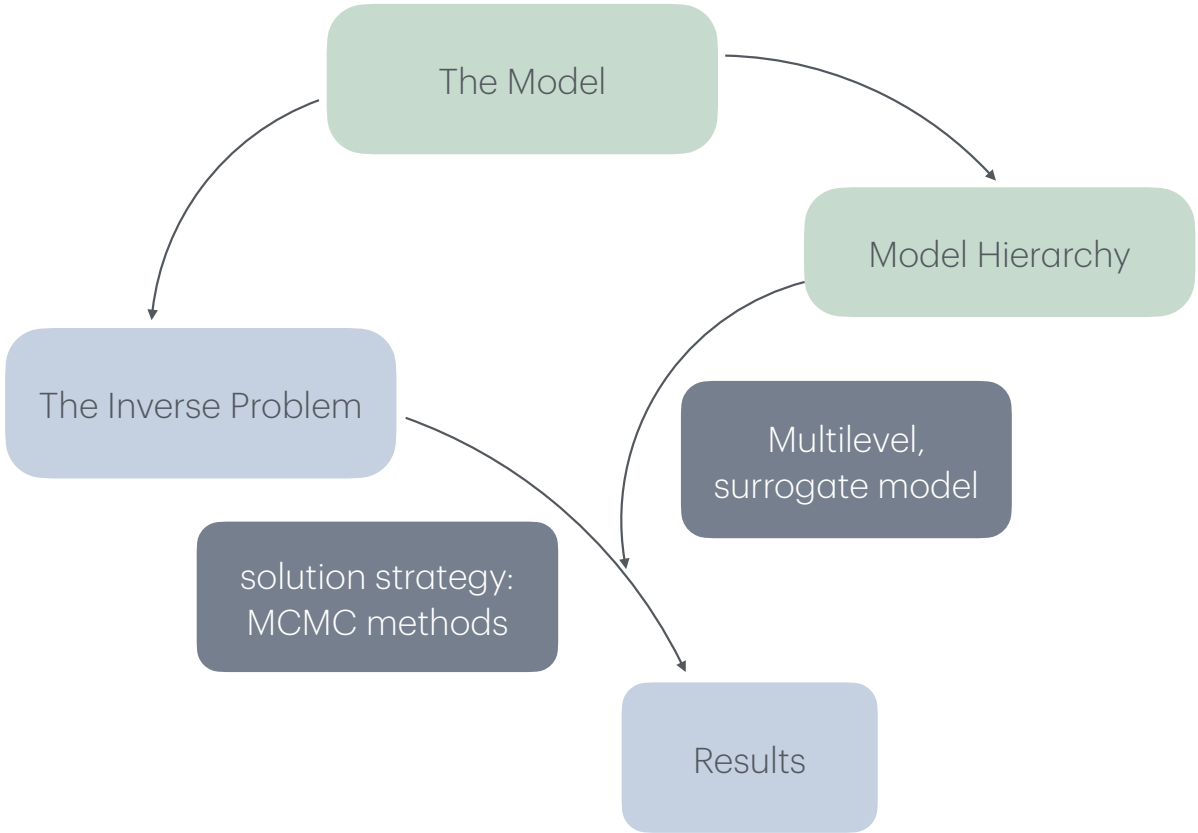
Source: TNG Collaboration



Pillepich et al.: Simulating galaxy formation with the IllustrisTNG model, 2017

The "Story"

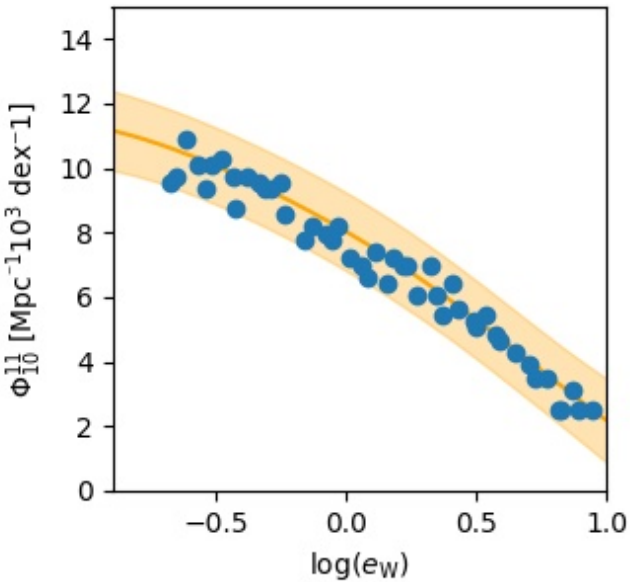
1. The Model/Hierarchy
2. The Inverse Problem:
   Model-data comparison
3. MLDA Setup and
   Surrogate Modeling
4. Results



The IllustrisTNG Model

from the perspective of parameter inference: stochastic behaviour

- more than 28 parameters, not all physical/ “measurable”
- some processes modelled are inherently stochastic
- model outputs to a (set of) parameters vary around a mean
- parameter influence on model outputs not clear, some parameters act as “dials” for stochasticity



No. of galaxies at certain mass range, depending on parameter e_W

The MLDA Algorithm (Lykkegaard et al. 2023)

"Hierarchical" Markov Chain Monte Carlo

- sampling from the posterior distribution $\pi(\theta \mid d) = \dfrac{\pi(d \mid \theta)\pi_p(\theta)}{\pi(d)}$

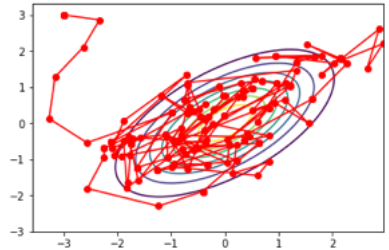
  of parameter $\theta \in \mathbb{R}^R$, given uncertain measurements $d \in \mathbb{R}^D$

- Delayed Acceptance: approximate (low-res) model evaluations and/or surrogates to accelerate Metropolis-Hastings by improving proposals

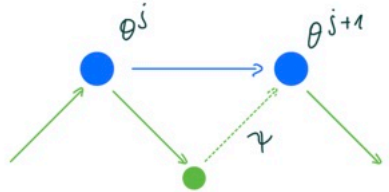
- Mutilevel Delayed Acceptance: recursively calling delayed acceptance with subchains of arbitrary length

Crucial: convergence to target distribution is maintained.

Source: Rel Guzman



Metropolis-Hastings



Delayed Acceptance



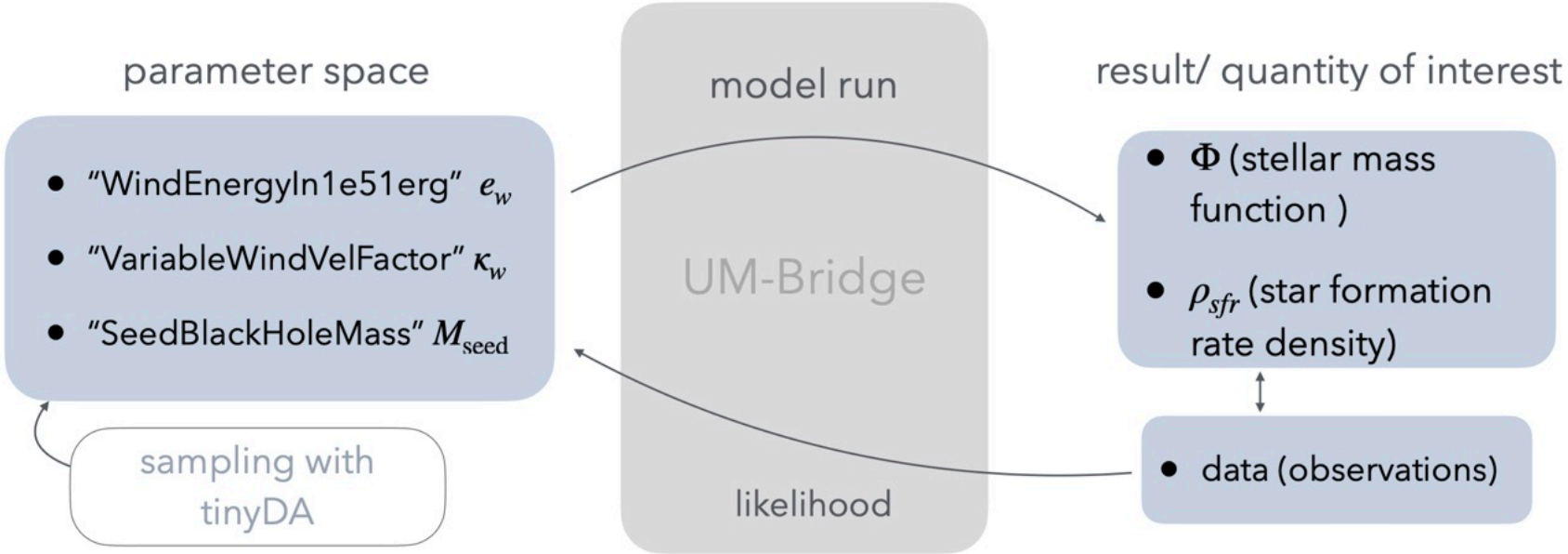
Multilevel Delayed Acceptance

Running MLDA

(Seelinger et al. 2025)

HPC Setup

- challenge: connecting MLDA sampling code to (extremely expensive) model evaluations



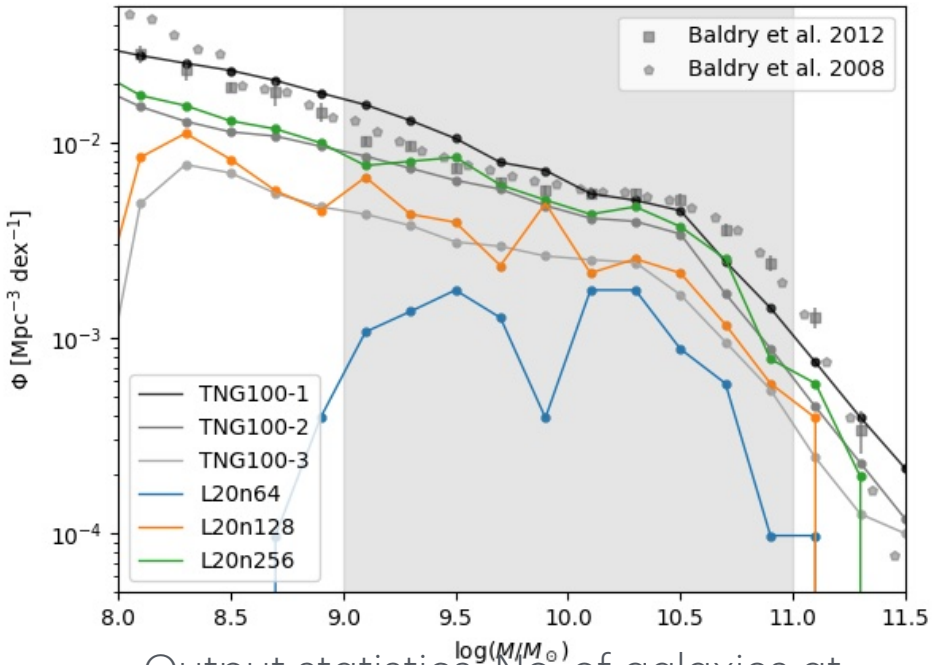
Model Hierarchy

Level structure through varying numerical resolution

- solve “same” initial matter distributions with different degrees of numerical resolution
- all (except for smoothing-related) parameters kept unchanged
- observe: model behaviour changes with resolution

model config.	cpu hours (est.)
L20n64	32-50
L20n128	512-1.800
L20n256	4.000-13.000

CPU hours for model evaluations (parameter dep.)

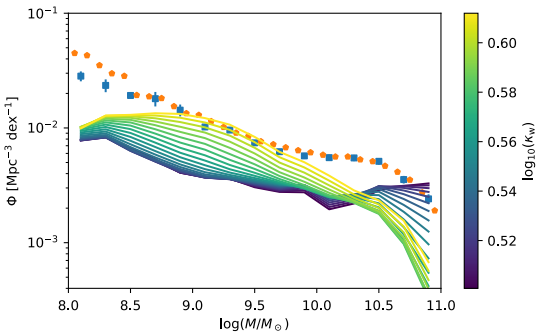


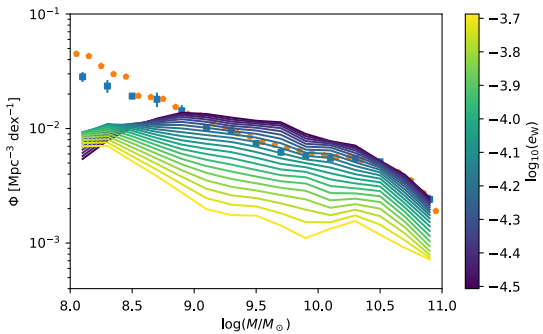
Output statistics: No. of galaxies at certain galaxy mass vs. galaxy mass

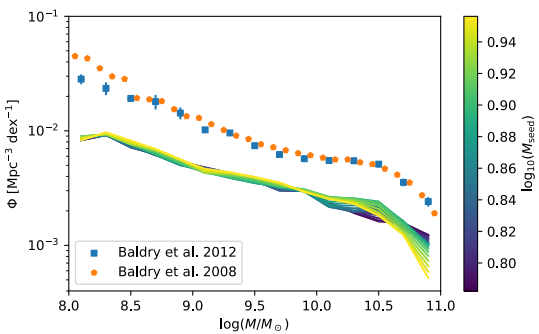
Surrogate Modeling

Emulating the model through Gaussian process regression

- Gaussian process regression on 600 (Latin hypercube) sample evaluations: Matern-Kernel + White Noise
- emulate the parameter-dependance of the Galaxy stellar mass function (output statistics of TNG simulations)
- “meaningful" parameter dependance: noise fitting successful
- model evaluations in less than 1 second!







Running MLDA

Results

- corner plot: running chains of 100.000 steps on the surrogate model
- result: expectation of the posterior differs from “fiducial” parameters
- Remark: data quality not quite high enough for inference, Gaussianity might be a too strong assumption

Have shown: Parameter inference for extremely expensive and partially stochastic simulations!

And: Similar methods can be applied for other large computational models... maybe yours?

Table: parameter space/ priors

parameter	fiducial value	range
e_w	3.6	0.1 – 10
κ_w	7.4	0.74 – 74
M_{seed} [$10^{10} M_\odot$]	0.00008	10^{-6} – 10^{-3}

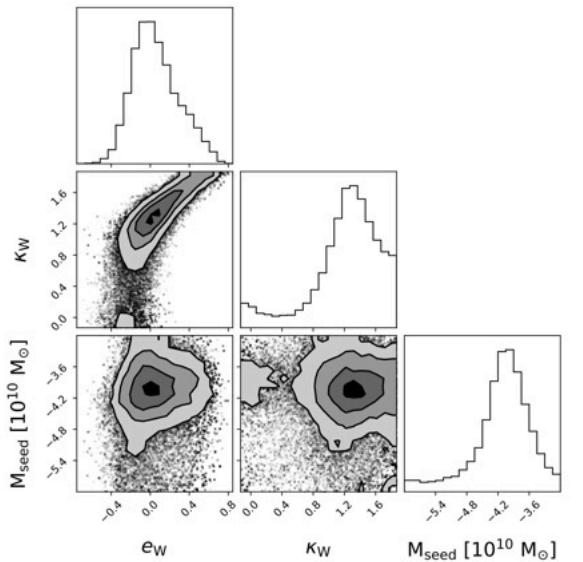


Figure: empirical posterior from surrogate.

NOTHING TO SEE HERE.

From a Mathematical PoV

Is this inverse problem ill-posed?

- non-linear measurable "forward map" $\mathcal{F} : \mathbb{R}^R \to \mathbb{R}^M$
- corresponding "forward problem": compute $\mathcal{F}(\theta)$ for some $\theta \in \mathbb{R}^R$
- "inverse problem": for given observations $d \in \mathbb{R}^D$, find θ such that $\mathcal{G}(\mathcal{F}(\theta)) = d$
  - in reality $d = \mathcal{G}(\mathcal{F}(\theta)) + \varepsilon + \eta = \mathcal{G}(\mathcal{F}(\theta)) + \tilde{\varepsilon}$
  - we make the assumption $\tilde{\varepsilon} \sim \mathcal{N}(0, \Sigma_{\tilde{\varepsilon}})$
  - covariance matrix $\Sigma_{\tilde{\varepsilon}}$